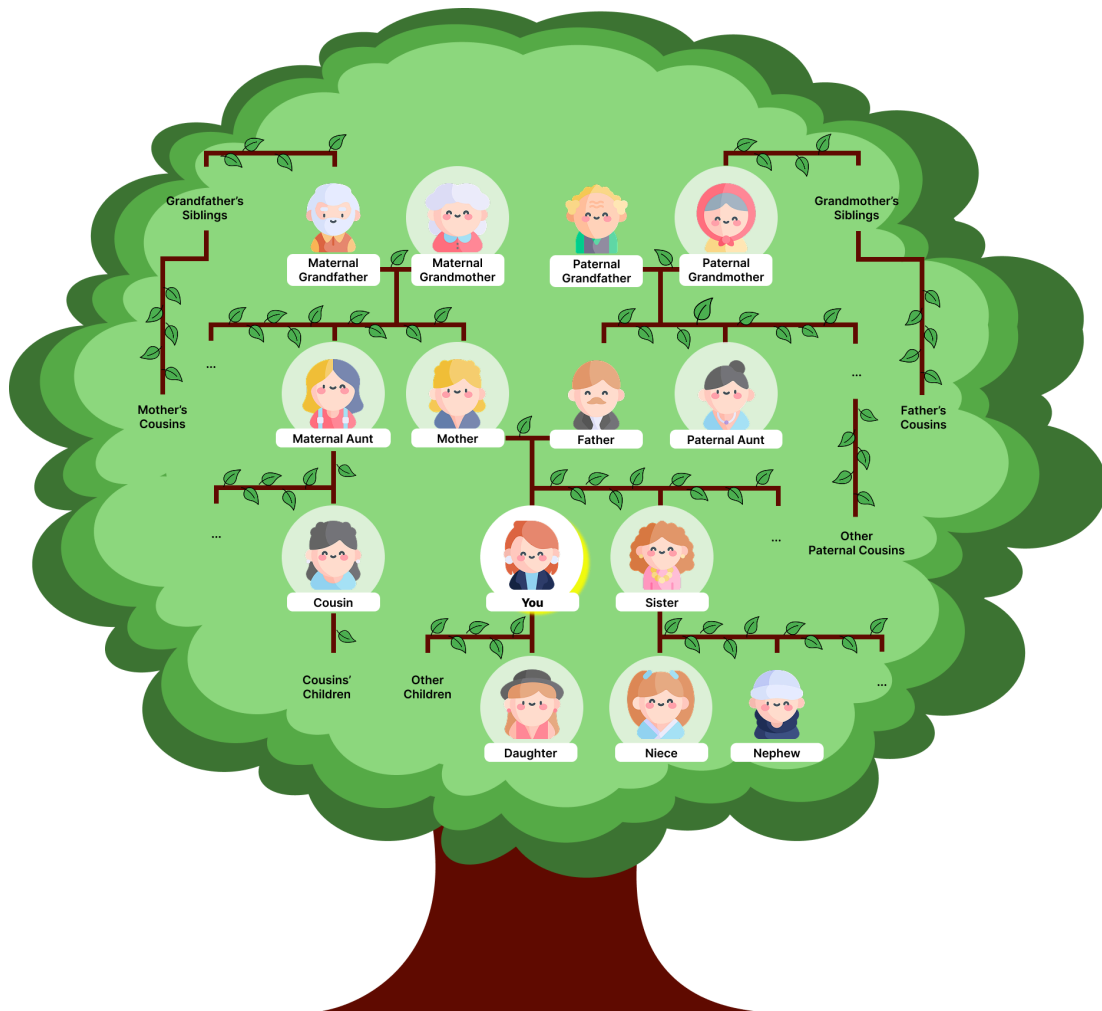


Introduction

This survey explores how women's health experiences are shared and learned within families. Women's health can be shaped by genetic factors, family history, and intergenerational knowledge. By understanding these experiences, we can gain insights into communication patterns, barriers, and opportunities for support.



An example of a family tree. The circled members are women in the family. You may have more or fewer extended family members than what is shown in this graph.

This survey will take around 20 minutes. Your responses will help us learn more about these topics. We would love to hear stories about how you have known or shared about your family's women's health experiences, and your thoughts on technology's role in this process.

Thank you for your participation!

Section 0: Screening

1. **Age:** What is your age in years? (only include age ≥ 18)
 - a. (enter a number)
2. Approximately how many biological (genetic) family members are you in contact with¹ (including occasionally through family gatherings or social media)?
 - a. 1-3²
 - b. 4-6³
 - c. 7-10
 - d. 11-20
 - e. 21-30
 - f. 31-40
 - g. 41-65
 - h. More than 65
3. Among your biological (genetic) family members, whose health conditions do you know about or have heard of, either directly or through others? (Select all that apply)
 - a. Mother
 - b. Father
 - c. Sister(s)
 - d. Brother(s)
 - e. Cousin(s)
 - f. Paternal grandmother
 - g. Paternal grandfather
 - h. Paternal aunt(s)
 - i. Paternal uncle(s)
 - j. Maternal grandmother
 - k. Maternal grandfather
 - l. Maternal aunt(s)
 - m. Maternal uncle(s)
 - n. Grandparents' siblings
 - o. Parents' cousins
 - p. Others (please specify)
4. How many of your biological (genetic) family members do you discuss health-related topics with (such as sharing personal health experiences or family health history)?
 - a. 1-3
 - b. 4-6
 - c. 7-10
 - d. 11-20
 - e. 21-30

¹https://www.researchgate.net/publication/263919053_Emotional_closeness_to_maternal_versus_paternal_lineages

² <https://www.census.gov/library/stories/2024/11/family-households.html>

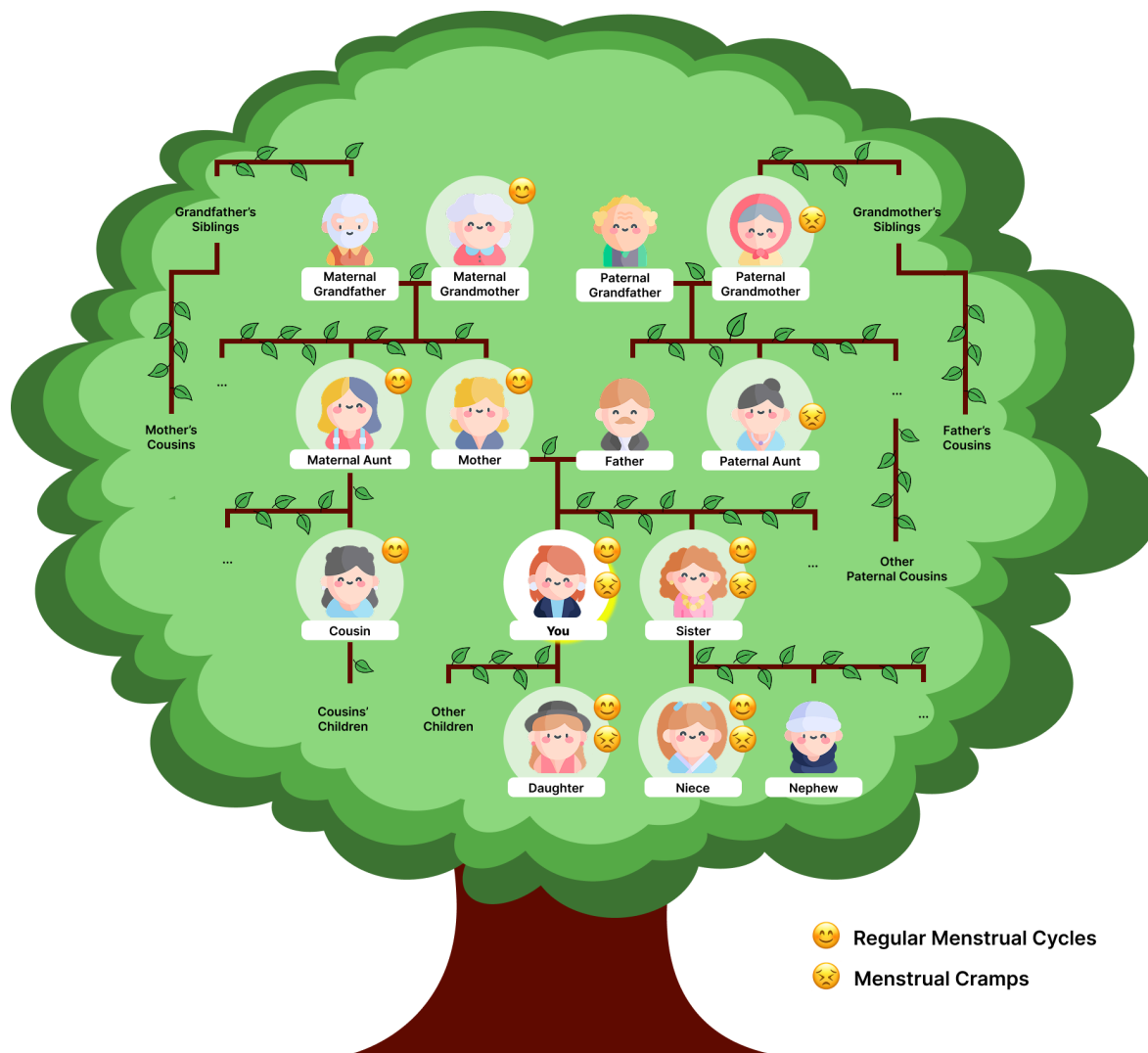
³ <https://pmc.ncbi.nlm.nih.gov/articles/PMC6711384/#S16>

- f. 31-40
 - g. 41-65
 - h. More than 65
- 5. Have you ever menstruated?
 - a. Yes
 - b. No
- 6. Have you ever given birth to a child?
 - a. Yes
 - b. No
- 7. Do you have one or more biological children who have menstruated?
 - a. Yes
 - b. No
- 8. Have you experienced *menopause* (the ceasing of menstruation)?
 - a. Yes, I am currently going through menopause
 - a. Yes, I have gone through menopause
 - b. No, I have not experienced menopause

Section 1: Awareness and Perception of Genetic Factors

Women's health includes (1) health issues that only affect women, like periods, pregnancy, and fertility conditions, and (2) health conditions that affect both men and women but may impact women differently, like heart disease and diabetes.⁴

Genetic factors in women's health refer to how inherited genes from family members can affect a woman's health. This can include things like the chances of getting certain illnesses or how the body handles changes like pregnancy or menopause.⁵



A family tree with annotations of women's health conditions. The "Regular Menstrual Cycles" and "Menstrual Cramps" are examples of women's health experiences that could be inherited across generations through genetics.

⁴ <https://www.nichd.nih.gov/health/topics/womenshealth>

⁵ <https://www.ncbi.nlm.nih.gov/books/NBK19932/>

8. **Knowledge:** How much do you think you know about how genetics affect women's health?
- a. Please rate from 1 to 5: Not at all knowledgeable (1) → Very knowledgeable (5)
 - b. Answer:
9. **Importance:** How important do you think it is to know about your family's health history, especially related to women's health issues?
- a. Please rate from 1 to 5: Not important (1) → Very important (5)
 - b. Answer:
10. **Impact:** How much do you think genetics affect women's health conditions?
- a. Please rate from 1 to 5: None (1) → A lot (5)
 - b. Answer:

Section 2A: Knowing about Family's Women's Health Experiences

- An example: You might be a young woman in active contact with your mother. You may **know** about your mother's menstrual pain because the two of you had some related conversations in the past. Your condition may vary from the example, but please **stick to your experiences** when answering the questions.
- Please also only consider your **biological** family and **women's health** aspects when you are answering the questions.

1. **Topics:** What aspects of your family's experiences related to women's health do you know about? (Check all that apply)
 - a. **Menstrual Patterns & Symptoms** (If your grandmother, mother or sister experiences painful cramps or irregular periods, you might have a higher chance of experiencing them too.⁶)
 - b. **Pregnancy & Fertility Experiences** (If women in your family have had pregnancy complications, or early labor, you might be more likely to face similar challenges.⁷)
 - c. **Menopause & Aging Changes** (If your mother went through menopause early or had severe symptoms like hot flashes, you might have a similar experience.⁸)
 - d. **Women's Illness or Chronic Conditions** (For example, if your mother or grandmother had breast cancer or cervical cancer, your own risk for these conditions may be higher.⁹)
 - e. **Body & Hormonal Changes Over Time** (If women in your family have experienced sudden weight gain, hair thinning, or energy changes, you might notice similar shifts as you age.)
 - f. Other: (please specify)
 - g. I do not know about my family's experiences related to women's health

(If a-f)

1. *Who did you learn about your family's women's health experiences from?* (Check all that apply)
 - a. Mother
 - b. Father
 - c. Grandparents
 - d. Aunts/Uncles
 - e. Siblings
 - f. Cousins

⁶<https://www.cambridge.org/core/journals/psychological-medicine/article/genetic-and-environmental-factors-in-the-aetiology-of-menstrual-premenstrual-and-neurotic-symptoms-a-populationbased-twin-study/D937E69809943136E5DECC44864F4BB4>

⁷ <https://pubmed.ncbi.nlm.nih.gov/23972758/>

⁸ <https://www.thieme-connect.com/products/ejournals/html/10.1055/s-0035-1567825>

⁹ <https://www.sciencedirect.com/science/article/pii/S1521693420300341>

- g. Others (specify)
- 2. **What do you know about your family's experiences related to women's health, and how did you find out?**

You can share examples like stories, including:

- a. When: Was there a specific occasion or situation?
- b. What: Whose experience was it? What was the topic, and how did it relate to your experiences? Could you share some details?
- c. How: Who did you know it from? Did you learn through conversations, observation, or technology (e.g., notes, health records)?
(Open-ended text entry, min 10 words)

(If not all of a-f have been selected)

- 3. **In your opinion, what might be the reasons or barriers that prevented you from learning about some of your family's women's health experiences?** If not applicable, please enter N/A.
 - a. (Open-ended text entry)

- 4. In the future, suppose you are able to know everything you want to know about your family's experiences about women's health. **What would you want to learn more about your family's women's health experiences?**
You may think about the family tree, and consider *whose experiences you are interested in, and when and how* you would like to learn these experiences.
(open-ended question)

Section 2B: Sharing Women's Health Experiences with Family

- An example: You might be a mother and have **shared** your struggles during pregnancy and childbirth with *your daughter or sister* in the past when they planned to have a child. You may have shared the experiences of your *other family members* as well. Your condition may vary from the example, but please **stick to your experiences** when answering the questions.
- Please also only consider your **biological** family and **women's health** aspects when you are answering the questions.

1. **Topics:** *What aspects* of you or your family's experiences related to women's health have you shared with your family? (Check all that apply)
 - a. **Menstrual Patterns & Symptoms** (If you experience painful cramps, heavy periods, or mood swings, your children might be more likely to have similar menstrual symptoms.)
 - b. **Pregnancy & Fertility Experiences** (If you had difficulty getting pregnant, pregnancy complications, or early labor, sharing your experience could help your children understand and prepare for potential challenges.)
 - c. **Menopause & Aging Changes** (If you went through menopause early or had severe symptoms like hot flashes and sleep issues, your children might want to know what to expect as they age.)
 - d. **Women's Illness or Chronic Conditions** (If you have a family history of breast cancer, heart disease, or osteoporosis, your children may be at higher risk and could benefit from knowing this.)
 - e. **Body & Hormonal Changes Over Time** (If you've experienced sudden weight gain, hair thinning, or energy changes, your children might notice similar shifts and want to understand how to manage them.)
 - f. Other: (please specify)
 - g. I have **not** shared my women's health experiences with my family to support them

(If a-f)

2. *Who* have you shared your or your family's women's health experiences with? (Check all that apply)
 - a. Mother
 - b. Father
 - c. Siblings
 - d. Cousins
 - e. Children
 - f. Grandchildren
 - g. Friends
 - h. Healthcare professionals
 - i. Others (specify)

5. **What have you shared about your family's women's health experiences with your family, and how did you share?**

You can share examples like stories, including:

- a. When: Was there a specific occasion or situation?
- b. What: Whose experience was it? About what topic and what details have you shared? Was it related to the experiences of other family members?
- c. How: Who did you share the experience with? Did you share through conversations, actions, or technology (e.g., notes, health records)?
(Open-ended text entry, min 10 words)

(If not all of a-e have been selected)

6. **What were the reasons or barriers that prevented you from sharing some of your women's health experiences with your family?** If not applicable, please enter N/A.

- a. (Open-ended text entry)

7. In the future, you may have new members in the family, or you or your family face new experiences related to women's health. **What would you like to share about your women's health experiences to your family then?** You may think about the family tree figure, and consider with whom, when and how you would like to share.

- a. (open-ended question)

Section 3: Technology Practice

1. What types of technology do you use to find or learn about health information? (Select all that apply)
 - a. Search engines and health websites (such as Google, Bing, WebMD, Mayo Clinic)
 - b. Generative AI tools (such as ChatGPT, Bard)
 - c. Health apps (such as MyChart, MedlinePlus, period or fitness trackers)
 - d. Social media or online communities (such as Facebook groups, health forums)
 - e. Communication with healthcare providers (such as patient portals, telemedicine)
 - f. Other (please specify)
 - g. I do not use technology for health information
2. What technologies have you used for recording or managing your personal health information related to *women's health*? (Check all that apply)
 - a. Health apps (for example, period or fertility trackers like Clue, Flo, fitness apps like MyFitnessPal)
 - b. Wearables (for example, Fitbit, Apple Watch, Oura Ring)
 - c. Manual tracking (for example, writing in a journal, calendar notes)
 - d. Medical records (for example, doctor's notes, hospital records, printed test results)
 - e. Other [please specify]
 - f. I have not used any of the technologies above
3. **How do you think the existing technology can be used or improved to help you and your family communicate about women's health experiences?** They could be the ones listed above or other technology that you can think of. If not applicable, please enter N/A.
 - a. (open-ended question)
4. In the future, **what other tools or digital technologies, if any, do you imagine could help you and your family communicate about women's health experiences?** You may consider the following examples and describe how they might work: physical tools like notes or photo books; health apps or tracking devices with long-term maintenance or sharing features; storytelling or AI technologies to support conversation.
(open-ended question)